CIT 140 · Week 7 · Data-Quality Report

Meridian Forge · fictional inspection export v1

Learner name / date	Jose Romero 11/6/2026
Import status	20 rows loaded

Amber cells: your responses. Other results calculate from Prepared and Audit.

Rule	Check	Rows flagged	Share of all rows	**Affected Row_IDs**
Q01	Machine ID is required	2	10%	R03, R14
Q02	Work order is unique	2	10%	R05, R06
Q03	Status uses the agreed list	2	10%	R08, R16
Q04	Duration is in range	3	15%	R10, R18, R20
Q05	READY identifies an inspector	3	15%	R03, R07, R12
Distinct rows with any issue		11	55%	Count each row once, even when several Q-rules flag it. Do not add the five counts.

Scope: all 20 supplied records. Q02 flags every row in a repeated Work_Order_ID group; extra duplicate copies are a different measure. Unresolved fields remain flagged, not guessed.

Source: authored CIT140 case W07-S01 (Raw) and W07-S02 (rules). Report a data-quality finding, not a machine safety clearance. Explain your evidence and next action on page 2.

Week 7 · Explanation and refresh evidence

Finding: Row_ID, Q-number, actual value, source and effect

One of the flagged rows was Row ID R03. The rules which R03 fails are Q01 and Q05. It failed Q01 because its Machine ID field is blank and it failed Q05 because its Inspector ID was blank. These problems matter to someone using the inspection list because they have to be marked for review as stated in W07-S02, guesses are not sufficent.

Next action: what to verify, who owns it, and how to retest

Since the Machine ID is blank, it needs to be checked by the Maintenance Records Owner and they should verify it against the original inspection record as stated in W07-S03. As for the blank Inspector ID which needs to be reviewed, the shift supervisor should look to verify it, supported by W07-S03. Following this, I would test rules Q01 and Q05 again.

Limit: why you cannot guess a correction or prove accuracy

One cannot guess a correction or prove accuracy because the inspection records help to explan what was recorded, they can't prove what happened to a real machine. For this reason, the problems must be marked for review.

Professional connection: one framework idea and how you used it

One of the framework ideas that I used from the data quality action plan was step 4, documenting your findings. This was highlighted from the way the affected Row IDs which failed Q01-Q05 were written down in order to be marked for review, and therefore allowing problems in the inspection records to be identified.

Refresh test: before, changed value, after, and restored result

Prior to the refresh test the Q04 count was 3. The inspection minutes value in Row ID R01 changed from 20 to 0. After this change the Q04 count became 4. Finally, restoring the inspection minutes value back to 20 restored the Q04 count back to 3.

Export only Report: two landscape pages, page break before row 26. Review both pages.